**FULL NAME:** ______________________ **BLOCK:** __________

# Solving Rational Equations + Chemistry DOK-3 (Q#2/Q#6/Q#7 Prep)

Lesson 4-5 · Quick · Teacher Edition

## OBJECTIVES

| | |
|---|---|
| **Math Objective** | Solve rational equations in one variable by clearing denominators, check for extraneous solutions, and apply rational equations to real-world rate and mixture problems. |
| **Language Objective** | Justify why a candidate solution is extraneous, using the domain restrictions of the original rational equation. |
| **Essential Understanding** | Multiplying both sides of a rational equation by an expression containing variables can introduce extraneous solutions — values that satisfy the cleared equation but violate the original domain. |

## TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

| | |
|---|---|
| **Topic Goals** | Fluency solving rational equations; identifying extraneous solutions; applying to work-rate and mixture problems (Topic 4 LEHS Q#2, Q#6, Q#7 prep). |
| **Essential Question** | When you solve a rational equation by clearing denominators, why must you check every candidate solution against the original equation? |
| **Materials** | Student packet, pencil, calculator. DOK-3 Practice #33 (chemistry mixture performance task) rehearses Topic 4 LEHS Q#6 (solve rational equation) and Q#7 (work-rate reasoning). |

### [PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 CAPSTONE

This is a single-period quick treatment of Lesson 4-5. TWO examples are modeled in Launch (deviation from standard Klimsara one-example rule — solving AND extraneous checking are both assessment-critical). Release promptly at minute 18.

Today's capstone is Practice #33 (chemistry alcohol mixture). Students must (a) set up $f(x) = (50 \cdot 0.02 + 0.06x)/(50 + x) = 0.05$, (b) clear the denominator and solve — $50 \cdot 0.02 + 0.06x = 0.05(50 + x) \to 1 + 0.06x = 2.5 + 0.05x \to 0.01x = 1.5 \to x = 150$ gallons, (c) explain calculator TABLE verification. Same algebraic move as Topic 4 LEHS Q#6; same reasoning as Q#7.

Pace: Try-It 1 (4 min) → Try-It 3 (4 min) → #10 (4 min) → #14 (4 min) → #25 (6 min) → #33 (10 min). ~32 min total Explore.

45-min Wed F variant: cut #14 + #25. Never cut #33.

*Tag: bridge from 4-4 to solving rational equations*

**Do Now** **DOK:** 1 **Minutes:** 5

**Teacher does**

- Hand out at door.
- Circulate 3 min.

- Cold-call 1 student on the conflicting candidate solutions.

**Students do**

- Read and decide whether either candidate is valid.
- Identify which (if any) is extraneous.

**Questions to ask**

- What does it mean for a candidate solution to be 'extraneous'?
- Why do we check candidates in the ORIGINAL equation, not the cleared one?

**Adult role**
Solo teacher: circulate silently. No hints.

**[MIC] BRIDGE SCRIPT (read aloud after Do Now)**

"Topic 4 so far: inverse variation, reciprocal functions, multiplying and adding rational expressions. Today: SOLVING rational EQUATIONS. Same algebra, but with an equation to solve and a new trap — extraneous solutions. The Model & Discuss shows two students solving a rational equation and getting different candidate solutions. Which (if any) is valid?"

### Do Now (4-5-savvas-model-discuss-lesson-4-5-launch)

**CRITIQUE & EXPLAIN**
Nicky and Tavon used different methods to solve the equation $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$.

| **Nicky** | **Tavon** |
|---|---|
| $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ | $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ |
| $\frac{1}{2}x = \frac{9}{10} - \frac{2}{5}$ | $10\left(\frac{1}{2}x + \frac{2}{5}\right) = 10\left(\frac{9}{10}\right)$ |
| $\frac{1}{2}x = \frac{5}{10}$ | $5x + 4 = 9$ |
| $x = 1$ | $5x = 5$ |
| The solution is 1. | $x = 1$ |
| | The solution is 1. |

**A.** Explain the different strategies that Nicky and Tavon used and the advantages or disadvantages of each.

**B.** Did Nicky use a correct method to solve the equation? Did Tavon?

**C.** Use Structure Why might Tavon have chosen to multiply both sides of the equation by 10? Could he have used another number? Explain.

**[CHECK] ANSWER KEY**

(verify)

*Tag: teacher models Example 1 + Example 3 [COMPRESSED — 2 examples]*

**Launch** **DOK:** 1–2 **Minutes:** 13

**Teacher does**

- Project Example 1 (solve rational equation). Think aloud: find LCD, multiply both sides, solve polynomial, check in original.
- "Every candidate must be checked in the ORIGINAL equation — clearing denominators can introduce values that don't belong."
- Transition to Example 3 without stopping — announce: "Now watch what happens when a candidate IS extraneous."
- Project Example 3. Think aloud: state domain restrictions FIRST, then solve; test each candidate against the restrictions.
- Flag the Rules card — students copy into margin.
- 30-sec pause: "Turn to partner: why does clearing denominators sometimes produce extraneous solutions? What's the mechanism?"
- Do NOT solve Try-Its or Practice items — release at minute 18.

**Students do**

- Watch Example 1 silently. Note the four-step procedure.
- Watch Example 3. Mark the domain-restriction identification step.
- During pause: explain mechanism to partner using Rule 1(d).
- Copy Rules card into margin.

**Questions to ask**

- What is the LCD in Example 1?
- After you clear denominators, what kind of equation do you get?
- In Example 3, what values are forbidden by the original denominators?
- Why does multiplying by an expression containing variables sometimes introduce extraneous roots?

**Adult role**
Project both examples back-to-back. Release promptly at minute 18.

**[MIC] TEACHER SCRIPT**

1. "Watch me solve a rational equation. Step 1: find the LCD."
2. "Step 2: Multiply BOTH SIDES by the LCD. This clears all fractions."
3. "Step 3: Solve the resulting polynomial equation — you'll get candidate solutions."
4. "Step 4: CHECK each candidate in the ORIGINAL equation. Any candidate that makes a denominator zero is extraneous — throw it out."
5. "Example 3: watch what happens when a candidate IS extraneous. I state the domain restrictions FIRST — before I even start solving — so I know what's forbidden."
6. "Today's DOK-3 task is a chemistry alcohol mixture. Same procedure — set up rational equation,

clear, solve, check. Same algebraic move as Topic 4 LEHS Q#6."

7. Release to Explore.

*Tag: TPS + DOK-3 driver*

**Explore** **DOK:** 2–3 **Minutes:** 32

**Teacher does**

- 5 laps across 32 min.
- Lap 1 (4 min): Try-It 1 — solve. Press pairs on the LCD identification.
- Lap 2 (4 min): Try-It 3 — extraneous. Watch for students who skip the check.
- Lap 3 (4 min): #10 — solve + check. Confirm the check step.
- Lap 4 (4 min): #14 — extraneous trap. Press "which candidate violates the original domain?" on every pair.
- Lap 5 (6 min): #25 — work-rate. Use the $1/a + 1/b = 1/t$ setup.
- Lap 6 (10 min): [STAR] #33 Chemistry. Students set up the mixture equation, clear, solve for $x$, and explain TABLE verification.
- Do NOT give #33 Part A numeric answer. Pairs present argument at Share.

**Students do**

- 6 items in order: Try-It 1 → Try-It 3 → #10 → #14 → #25 → #33.
- For #33: BEFORE computing, write out $f(x) = (50 \cdot 0.02 + 0.06x)/(50 + x)$ and set equal to 0.05. Ask: what would $x$ mean in gallons?
- Justify TABLE verification with sentence frame.

**Questions to ask**

- Try-It 1: what's the LCD of the fractions?
- Try-It 3: which candidate makes the original denominator zero?
- #10: after you solve, WHY do you still need to check?
- #14: state the domain restrictions BEFORE you clear denominators.
- #25: how does $1/a + 1/b = 1/t$ capture the work-rate idea?
- #33 Part A: set up the mixture equation. What does $x$ represent in gallons — does your answer make sense?
- #33 Part B: how do you use $y_1$ and $y_2$ in the TABLE to verify?

**Adult role**
Solo teacher: 6 laps. On #33, press the mixture-equation setup AND the gallons reasonableness check — those are the assessment-critical moves.

**Try It 1 (4-5-savvas-try-it-1-lesson-4-5)**
What is the solution to each equation?

**a.** $\frac{2}{x+5} = 4$

**b.** $\frac{1}{x-7} = 2$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): $x = -4.5$

**Try It 3 (4-5-savvas-try-it-3-lesson-4-5)**

What is the solution to the equation $\frac{1}{x+2} + \frac{1}{x-2} = \frac{4}{(x+2)(x-2)}$?

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): no solution

**Practice #10 (4-5-savvas-q10)**

Error Analysis Describe and correct the error Miranda made in solving the rational equation.

$$\frac{1}{x-2} + \frac{x-2}{x+2} = \frac{x-4}{x-2}$$
$$(x+2)(x-2)\left(\frac{1}{x-2} + \frac{x-2}{x+2}\right) = \left(\frac{x-4}{x-2}\right)(x+2)(x-2)$$
$$(x+2)(1) + (x-2)(x-2) = (x+4)(x+2)$$
$$x + 2 + x^2 - 4x + 4 = x^2 + 6x + 8$$
$$-3x + 6 = 6x + 8$$
$$-2 = 9x;\ \text{or } x = -\frac{2}{9}\ \text{X}$$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): Miranda incorrectly expanded the right side as $(x+4)(x+2)$ instead of $(x-4)(x+2)$. The correct right side is $x^2 - 2x - 8$. Solving $-3x + 6 = -2x - 8$ yields $x = 14$.

**Practice #14 (4-5-savvas-q14)**

Make Sense and Persevere Solve the rational equation shown. Explain what is unique about the solution.

$$\frac{x^2 - 7x - 18}{x+2} = x - 9$$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): The solution is all real numbers except $x = -2$. Factoring the numerator shows that the equation is an identity $((x-9)(x+2))/(x+2) = x - 9$, which is true for all $x$ in the domain.

**Practice #25 (4-5-savvas-q25)**

Make Sense and Persevere Kenji can finish a puzzle in 2 hours working alone. Oscar can finish the same puzzle in 3 hours working alone. How long would it take Oscar and Kenji to finish the puzzle if they worked on it together?

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): 1.2 hours

**[SPEAK] CHAIN 2 CALLBACK — Chemist Mixture (DE-FACTO PLANT, first rendered item)**

*When to say:* Before students start Practice #33. L35_P3's q30 (Venetta) was not rendered in that packet, so Chemist becomes the first student-visible Chain 2 item. Plant the archetype here.

"Chain 2 is about **operating a given model**. Today the chemist hands us a rational equation — the one that relates alcohol percentage to gallons added — and we solve for a specific amount. Whereas Chain 1 (Storage Box, Wavelength) asked us to **build** the model from a constraint, here the equation is **given**. We operate it. Same move will appear in Half-Life (Topic 5) and Compound Interest (Topic 6) — different function families each time."

*Tip:* the extraneous-solution check at the end is shared with Chain 5 (extraneous chain) — side-note it aloud.

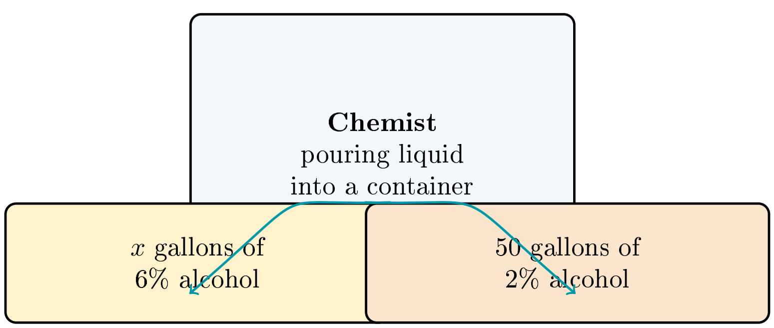
### Practice #33 [STAR] DOK-3 CHEMISTRY MIXTURE (4-5-savvas-q33)

**Performance Task**

A chemist needs alcohol solution in the correct concentration for her experiment. She adds a 6% alcohol solution to 50 gallons of solution that is 2% alcohol. The function that represents the percent of alcohol in the resulting solution is

$$f(x) = \frac{50(0.02) + x(0.06)}{50 + x},$$

where $x$ is the amount of 6% solution added.




*[IMAGE: Chemist pouring liquid into a container; labels "x gallons of 6% alcohol" and "50 gallons of 2% alcohol."]*

**Part A** How much 6% solution should be added to create a solution that is 5% alcohol?

**Part B** Use Appropriate Tools Explain the steps you could take to use your graphing calculator to verify the correctness of your answer to part (A).

**[STAR] DOK-3 CHEMISTRY MIXTURE — Practice #33 (Topic 4 LEHS Q#6 + Q#7 prep)**

q33 is the chemistry alcohol-mixture performance task: 50 gal of 2% alcohol solution, add $x$ gal of 6% alcohol, resulting concentration $f(x) = (50 \cdot 0.02 + 0.06x)/(50 + x)$. Part A: solve $f(x) = 0.05$ for $x$. Part B: explain how to verify with a graphing calculator table.
"The move Part A wants is: set up the rational equation, clear the denominator, solve the resulting linear equation for $x$. Students should recognize this as a rational equation in disguise — same moves as Ex 1."
For Part B: enter $y_1$ = numerator, $y_2 = 0.05(50 + x)$, use TABLE to find where they intersect.
If stuck: ask 'what would the answer $x$ mean in gallons?' before they compute, to anchor the reasonableness check.
Answer: 150 gallons.
Sentence frame required for ELL.

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source): Part A 150 gallons. Part B Using the graphing function of your calculator, enter $2.5 + 0.05x$ for $y_1$, and $1 + 0.06x$ for $y_2$. Then use the TABLE function to find the $y$-values of the functions that are equivalent when $x = 150$. • IMAGE PLACEHOLDER [illustration]: Chemist pouring liquid into a container; labels "x gallons of 6% alcohol" and "50 gallons of 2% alcohol."

*Tag: academic conversation + Topic 4 LEHS Q#2/Q#6/Q#7 bridge*

**Share / Summary** **DOK:** 2 **Minutes:** 5

**Teacher does**

- Cold-call 1 pair to present their chemistry solution using sentence frame.
- Class signals thumbs. Write $x = 150$ gallons on board.
- Topic 4 bridge: "This is the exact move on Topic 4 LEHS Q#6, the domain reasoning is Q#2, and the rate reasoning is Q#7. You have now rehearsed all three."

**Students do**

- Presenting pair reads their solution and cites the TABLE verification.
- Rest of class signals and writes takeaway in margin.

**Questions to ask**

- Summarize: how did you get from $f(x) = 0.05$ to $x = 150$?
- Why did 150 gallons make sense as an answer?

**Adult role**

Cold-call a pair that struggled on #33 — hold them accountable to the setup step.

*Tag: summary recap (not graded)*

**Exit Ticket** **DOK:** 1–2 **Minutes:** 3

**Teacher does**

- Collect at door.

- Scan stem 1 (original / extraneous). Plan re-teach if many students can't articulate WHY the check is required.

**Students do**

- 3 sentence stems, honest.

**Questions to ask**

- Did students connect the 'check in original' step to avoiding extraneous solutions?
- Did students correctly describe the mixture-equation setup?

**Adult role**
Collect. No grade.

**[CLIPBOARD] EXIT TICKET — Student Form**

After clearing denominators in a rational equation, I must check each candidate solution in the _____ equation to catch _____ solutions.
In Practice #33 (chemistry), the equation came from setting _____ equal to the target concentration 0.05.
One biggest thing I learned today: ______________________________

### [CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for this lesson. Use the extra 10 min on Explore #33 (Chemistry) — richer reasonableness and TABLE-verification talk pays off on Topic 4 LEHS Q#6/Q#7.
45-min Wed F variant: cut #14 + #25 from Explore. Never cut #33.
If finished early: hand out Practice #8 (work-rate validity reasoning) as fast-finisher; rehearses Q#7 specifically.

### [PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Solving Rational Equations Rules card as a sticker/cut-out.
- For #33, provide the setup: "$f(x) = (50 \cdot 0.02 + 0.06x)/(50 + x) = 0.05$. Multiply both sides by $(50 + x)$." Students solve from there.
- Accept verbal TABLE-verification explanation in lieu of written steps.

### [GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "extraneous" = an answer that looks right algebraically but doesn't actually work in the original equation; "LCD" = least common denominator; "domain restriction" = a value that $x$ is not allowed to be.
- "Concentration" = amount of substance per unit of mixture; "mixture" = two solutions combined.
- Pair ELL students with English-dominant partners for TPS.